BEFORE YOU READ

Every family leaves a shape

Graphs of the common function families fall into a small set of recognisable curves.

L0 **Types of curves** by function family

- L1 **Line** constant slope, no curvature — $mx + b$
- L1 **Parabola** single vertex; opens up if a > 0 — $ax^2 + bx + c$
- L1 **S-curve** one inflection point — $ax^3 + \ldots$
- L1 **Hyperbola** two branches, asymptotes — $1/x$
- L1 **Half-curve** starts at a point, goes one way — $\sqrt{x}$
- L1 **V-shape** two linear pieces meeting at a vertex — $|x|$
- L1 **Wave** periodic, bounded oscillation — $\sin, \cos$
- L1 **Growth or decay** stays on one side of the x-axis — a^x
- L1 **Staircase** horizontal segments with jumps — step functions

TAKEAWAY

Read the family from the curve

See the function families page for each family in detail.